

PIYUSH SHAH

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SUMMARY

Instrumentation and Control Engineering Technology student with hands-on experience in industrial instrumentation, control systems, and sensor integration. Proficient in PLC programming, 4–20 mA instrumentation loops, sensor calibration, and troubleshooting. Experienced working with temperature measurement and control systems. Seeking a Summer 2026 co-op opportunity to apply my skills in control systems, instrumentation, and troubleshooting in a hands-on technical role.

EDUCATION

Instrumentation and Control Engineering Technology

Fall 2024 – Present

Red River College

- Coursework includes process measurement, control systems, PLCs, power distribution, industrial instrumentation, schematic design
- Extensive lab experience with sensor calibration, 4-20 mA signal loops, PLC programming, troubleshooting, control loop tuning, and HMI integration

Bachelor of Computer Science, (coursework completed)

Fall 2022 – Spring 2024

University of Manitoba

- Completed two years of university coursework in software development, algorithms, and data structures
- Developed strong problem solving and data analysis skills using C++, Python, and Java

TECHNICAL SKILLS

Electrical & Wiring: Control panel assembly, schematic interpretation, wire routing/termination, soldering, connector assembly, functional testing

Instrumentation & Controls: 4-20 mA loops, sensor/transmitter calibration, PID tuning, PLC ladder logic, LabVIEW HMI, OPC UA communication

Automation & Data Acquisition: Arduino-based monitoring systems, sensor integration, signal conditioning, real-time data acquisition, vibration, and current monitoring

Programming & Software: Arduino, C++, Python, MATLAB, Multisim, Solid Edge, AutoCAD, Visio, Excel, Word, PowerPoint

Documentation: Schematics, test reports, technical manuals, logbook

ACADEMIC & TECHNICAL PROJECTS

AI-Assisted Predictive Maintenance System | *Personal Project*

Dec 2025 – Feb 2026

- Designed and implemented a predictive maintenance system to monitor motor health using vibration, temperature, and current measurements
- Integrated accelerometer, temperature sensor, and current sensor with Arduino for real-time data acquisition
- Logged operational data and applied machine-learning techniques to classify healthy versus failing motor conditions
- Implemented threshold-based alarms and emergency shutdown logic
- Visualized live sensor data and fault indicators using a local SCADA-style monitoring dashboard.

Temperature Monitoring & Control System

Jan 2026

- Developed a closed-loop temperature monitoring system using thermistors and analog signal conditioning
- Implemented proportional and PID control logic to regulate system temperature
- Monitored process variables through real-time data acquisition
- Implemented threshold-based alarm logic to simulate overheating or system faults
- Analyzed system response and optimized controller parameters for stable operation

RTD Temperature Transmitter Project

Oct 2025 – Nov 2025

- Designed and simulated an RTD-based temperature transmitter that converts small resistance changes into a standard 4-20 mA industrial current signal.
- Used an instrumentation amplifier to accurately amplify the low-level differential signal from the RTD network, then applied additional signal conditioning to set the required gain and offset.
- Implemented a transistor-based current loop driver with a 250 Ω resistor to generate and verify correct loop current under load.
- Checked voltage and current at multiple test points throughout the circuit to confirm correct operation.

WORK EXPERIENCE

Electrical Technician

Tamarack Industries

June 2025 – Aug 2025

Winnipeg, Manitoba

- Assembled and wired 50+ industrial control panels and chassis harnesses for glycol heating equipment, interpreting schematic drawings to ensure compliance with client specifications and industrial standards
- Prepared layout boards for engine chassis harnesses and organized wiring routes prior to assembly.
- Performed wire cutting, routing, crimping, and connector termination. Conducted rigorous functional testing and inspections, achieving near-zero defects in completed units
- Collaborated with engineers to troubleshoot production issues, minimizing downtime, and enhancing assembly efficiency

Cook

Fairmont Hotel and Bull & Filly

Aug 2024 – Present

Winnipeg, Manitoba

- Developed strong attention to detail when preparing orders, ensuring consistency and accuracy.
- Worked as part of a team in a fast-paced environment, improving communication and coordination skills
- Learned how to work efficiently under pressure while keeping tasks organized and on schedule.

KEY STRENGTHS & SOFT SKILLS

- Strong interest in automation, instrumentation, and building control systems
- Detail-oriented with strong troubleshooting and problem-solving abilities
- Hands-on learner who adapts quickly to new technologies and systems
- Effective communicator and team contributor
- Organized and capable of managing multiple technical tasks efficiently